

Landscape Architecture

PORT FOLIO





NETHMA JAYANETTI

Graduate Landscape Architect

My interest in landscape architecture is rooted in a long-standing connection with the natural world and a curiosity about how people can coexist with nature and its processes. I approach design through a conceptual and creative perspective, supported by ecological thinking and an understanding of people, culture, and place. My interests particularly focus on climate-responsive design, ecological restoration, resilient landscapes, and nature-based solutions. Through landscape architecture, I aim to explore thoughtful ways of protecting natural systems while creating meaningful and lasting places for people.

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Education

University of Moratuwa, Sri Lanka
Bachelor of Landscape Architecture
– CGPA: 3.75/4.20

Coursera (Online) 2024
Fundamentals of GIS (University of California)
BIM Fundamentals for Engineers (National Taiwan University)

ESOFT Metro Campus 2021-2022
Diploma in Information Technology
Diploma in English
Diploma in AutoCAD

Maliyadeva Balika Vidyalaya 2006-2019
Secondary Education

Work Experience

The Design Alliance (PVT)Ltd 2024 DEC-2026 JULY
Intern - Landscape Architect

Publications

FARU International Research Conference 2025
Faculty of Architecture Research Unit, 2025

Temporal evolution of blue - green infrastructure and perceived benefits:
a case study for urban context of Colombo, Sri Lanka

Achivements & Leadership

Best Presenter Awards by Theme of Blue-Green Infrastructure
FARU International Research Conference 2025

Students' Society of Landscape Architecture-University of Moratuwa
Secretary 2026
Vice Secretary 2025

Publications

FARU International Research Conference 2025
Faculty of Architecture Research Unit, 2025

Temporal evolution of blue - green infrastructure and perceived benefits:
a case study for urban context of Colombo, Sri Lanka

Software Skills

 Autodesk AutoCAD  SketchUp
 Autodesk Revit  Lumion
 Adobe Photoshop  Microsoft Office
 ArcGIS

Languages

English (Fluent)
Sinhala (Native)

C o n t e n t



Urban Pocket
Park

1-4



Cromostreet

5-8



Whispers of
Dreamscape

9-12



From Roots To
Radiance

13-16



Regrounding
the Landfill

1-4

Parks for people

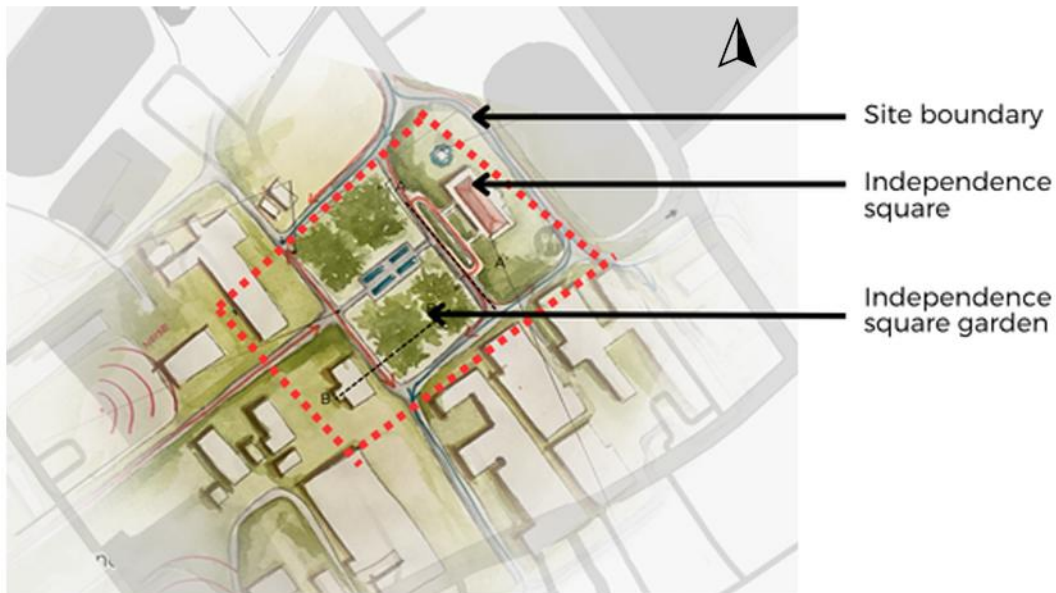
The design seeks to reduce the visual dominance of political representation through strategic landscape interventions, while respecting the existing Memorial Hall and heritage setting. By introducing a new people-centred focal axis and integrating existing landscape elements, the proposal creates a more inclusive public environment that reflects freedom, unity, reflection and shared experience, while supporting contemporary urban life and preserving Colombo's historical identity.



Context Analysis

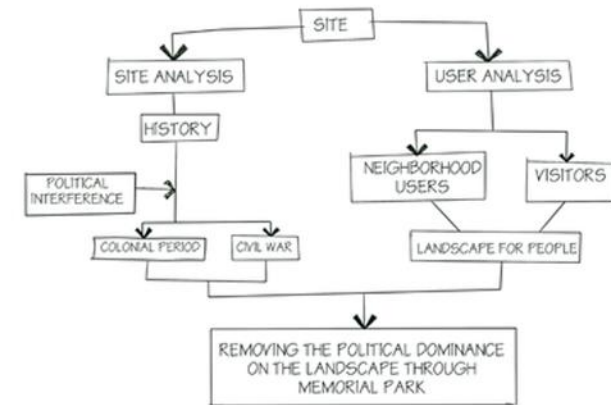


At the same time, the surrounding garden is actively used by people for walking, resting, recreation and social activity. This created an opportunity to shift attention from a purely monument-focused experience towards a stronger relationship between people, landscape and national identity.



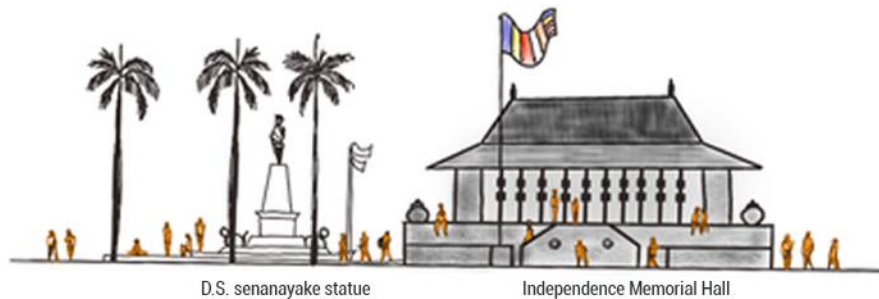
Site Analysis

Project Formulation

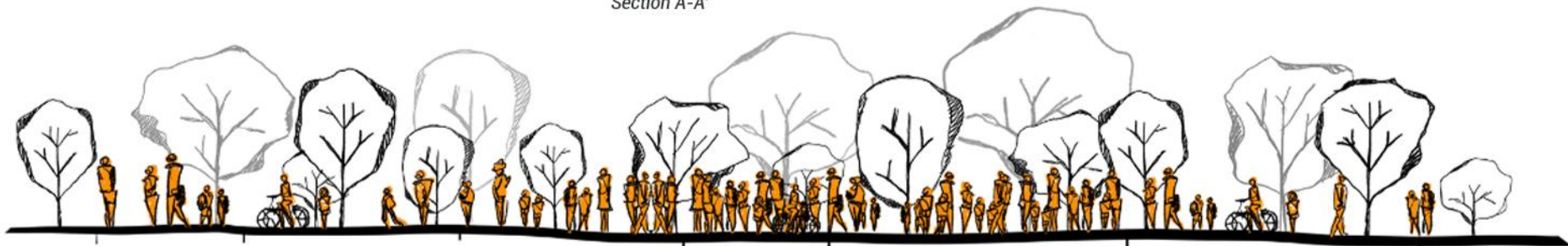


Independence Memorial Hall in Colombo, Sri Lanka, marks the nation's 1948 independence from British rule. Built at the site of the first self-rule ceremony, it symbolizes Sri Lanka's sovereignty and is a key cultural landmark, hosting national events and commemorations.

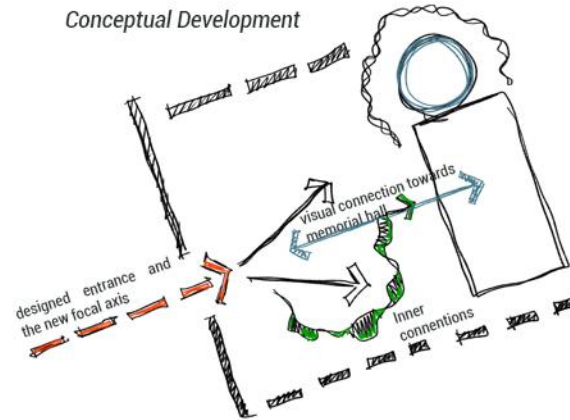
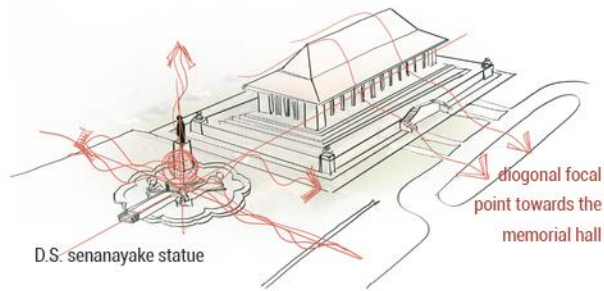
The statue of D.S. Senanayake the first prime minister of Sri Lanka, while intended to symbolize national unity and independence, also underscores the lasting influence of politics in shaping collective memory and its dominance on a cultural landscape.



Section A-A'



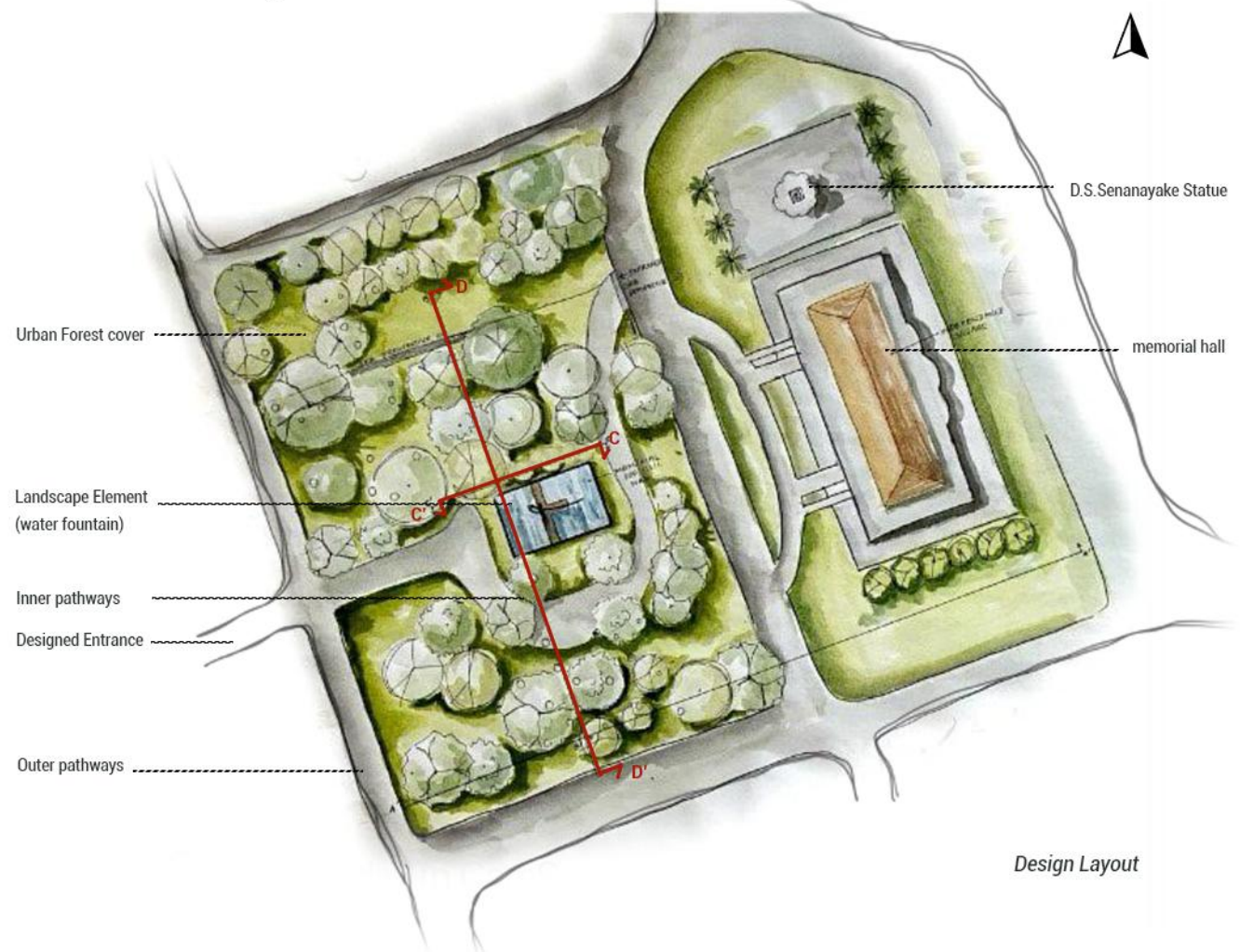
Section B - B'

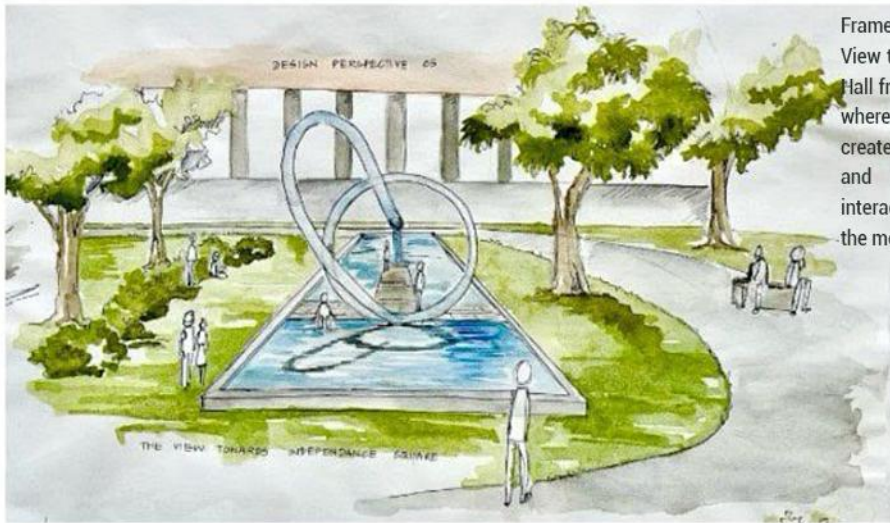


Replacing of political dominance with landscape elements that invite reflection, inclusivity, and a shared connection to the nation's heritage and future, transforming the space into a true symbol of unity and independence.

The conceptual sketches develop a new focal axis within the garden, drawing people towards a separate landscape experience rather than allowing the existing memorial composition to remain the only dominant visual frame.

The new focal space becomes a setting for reflection, gathering and shared experience, supporting the project's aim of strengthening inclusivity and a wider connection to independence and national heritage.

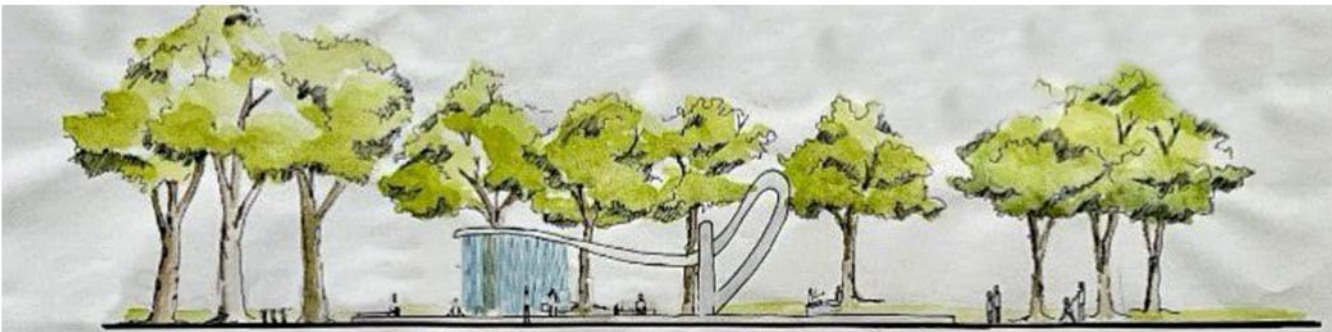




Framed Arrival
View towards the Memorial Hall from the new entrance, where the water feature creates a new focal layer and encourages public interaction before reaching the monument.

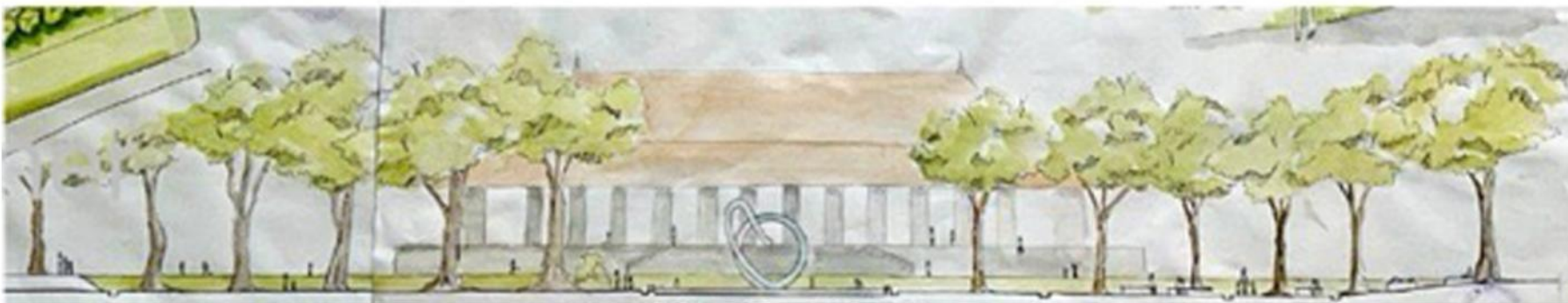


Urban Forest Walk
A shaded urban forest patch with internal pathways for walking, resting and outdoor activities, offering a calm green experience within the city core.

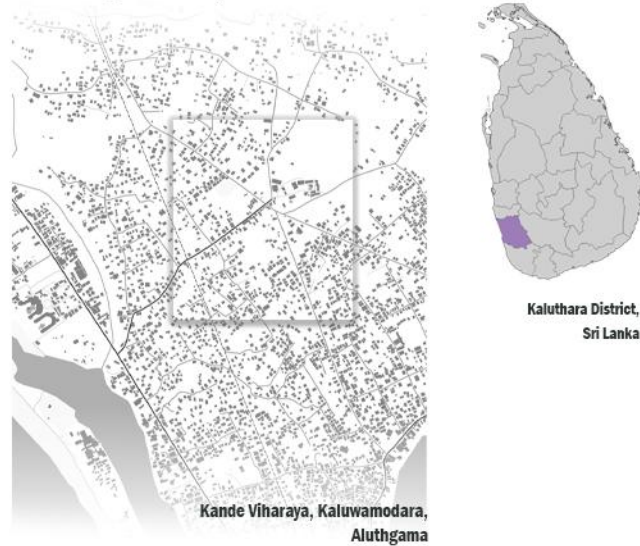


Design Section C - C'

Interactive Landmark
The sculptural form creates a distinctive public feature that supports interaction, gathering and memorable photographic moments within the landscape.



Design Section D - D'



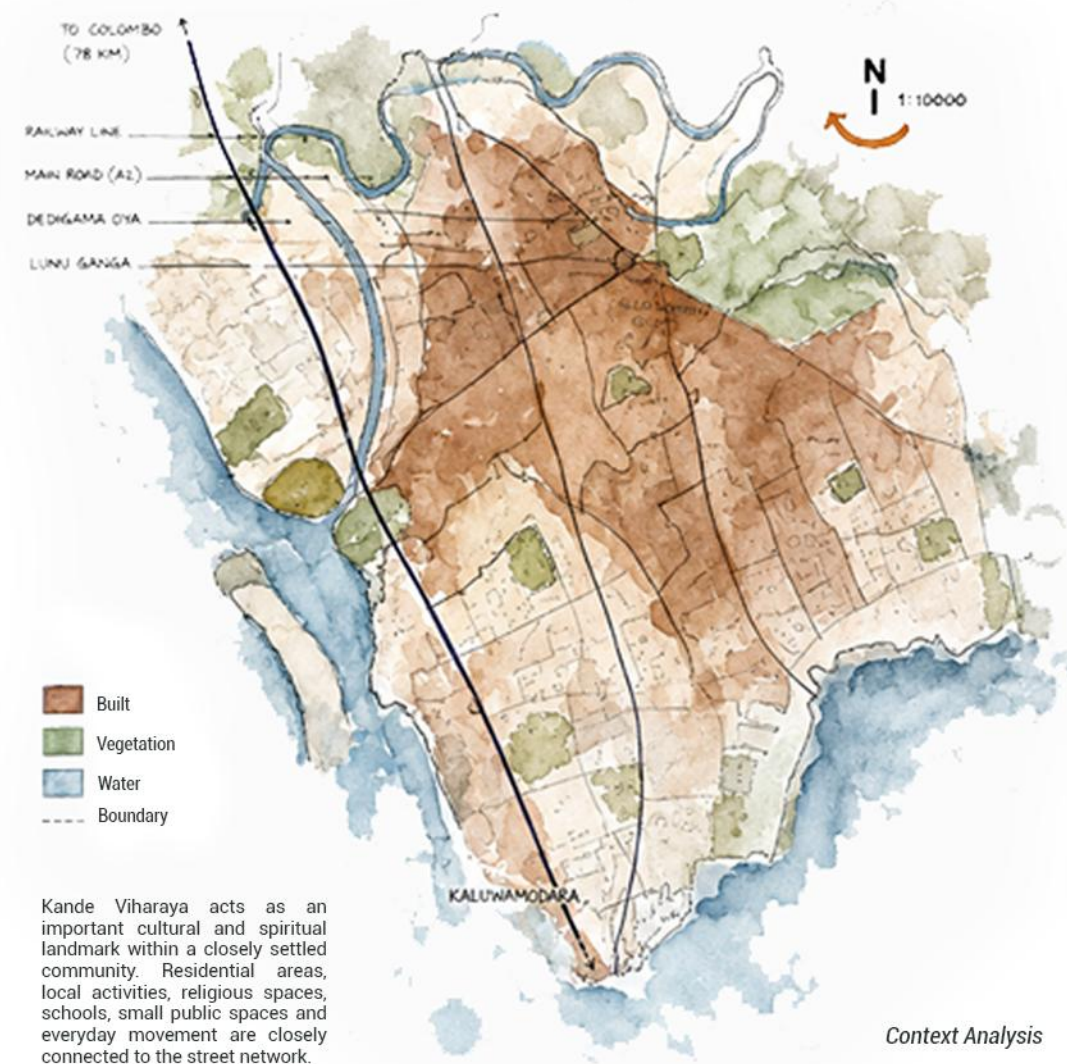
CHROMOSTREET

Reconnecting Community, Culture and Sacred Arrival

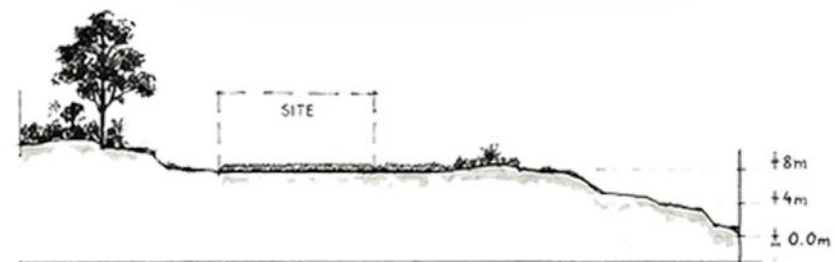
Located within the community surrounding Kande Viharaya, Kaluwamodara, Aluthgama, the project explores how a street functions not only as an access route to the temple, but also as part of the everyday life of residents, visitors and pilgrims.

The project reimagines the street leading to Kande Viharaya as a people-centred landscape that supports both everyday community life and the cultural importance of the temple. Existing neighbourhood character is strengthened through safer pedestrian movement, shaded gathering spaces, planting, lighting and improved water management.

Cultural colours and landscape elements create a gradual sense of arrival, while small community spaces encourage resting, meeting and interaction. Rather than separating the sacred destination from the surrounding settlement, the design uses landscape design to connect temple, street and community as one continuous cultural environment.



Kande Viharaya acts as an important cultural and spiritual landmark within a closely settled community. Residential areas, local activities, religious spaces, schools, small public spaces and everyday movement are closely connected to the street network.



Site positioning in the context

Key Issues Identified

Unsafe pedestrian movement – pedestrians and vehicles share limited street space.

Weak arrival identity – the route towards the temple does not clearly express its cultural importance.

Inactive spaces – several leftover and neglected spaces have little community use.

Limited shade and seating – few comfortable places are available for resting or informal gathering.

Poor continuity – pedestrian movement, planting and public spaces are fragmented.

Night-time limitations – insufficient lighting reduces comfort and safety.

Drainage issues – stormwater requires better management within the street landscape.

Existing Strengths

Kande Viharaya – a strong cultural landmark.

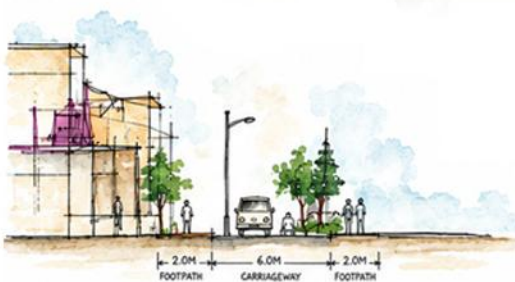
Established community – active neighbourhood relationships and daily street use.

Existing vegetation – mature trees and green spaces that can support shade and character.

Cultural elements – temple-related colours, walls, flags, spatial references and religious identity.

Existing public activity – pedestrian movement, gathering and visitor activity.

Unused spaces – opportunities for small community interventions.



Section P - P'

Site observation identified how people move, gather, wait, interact and approach the temple. Existing houses, vegetation, cemetery edges, religious elements and informal spaces already create a strong local character.

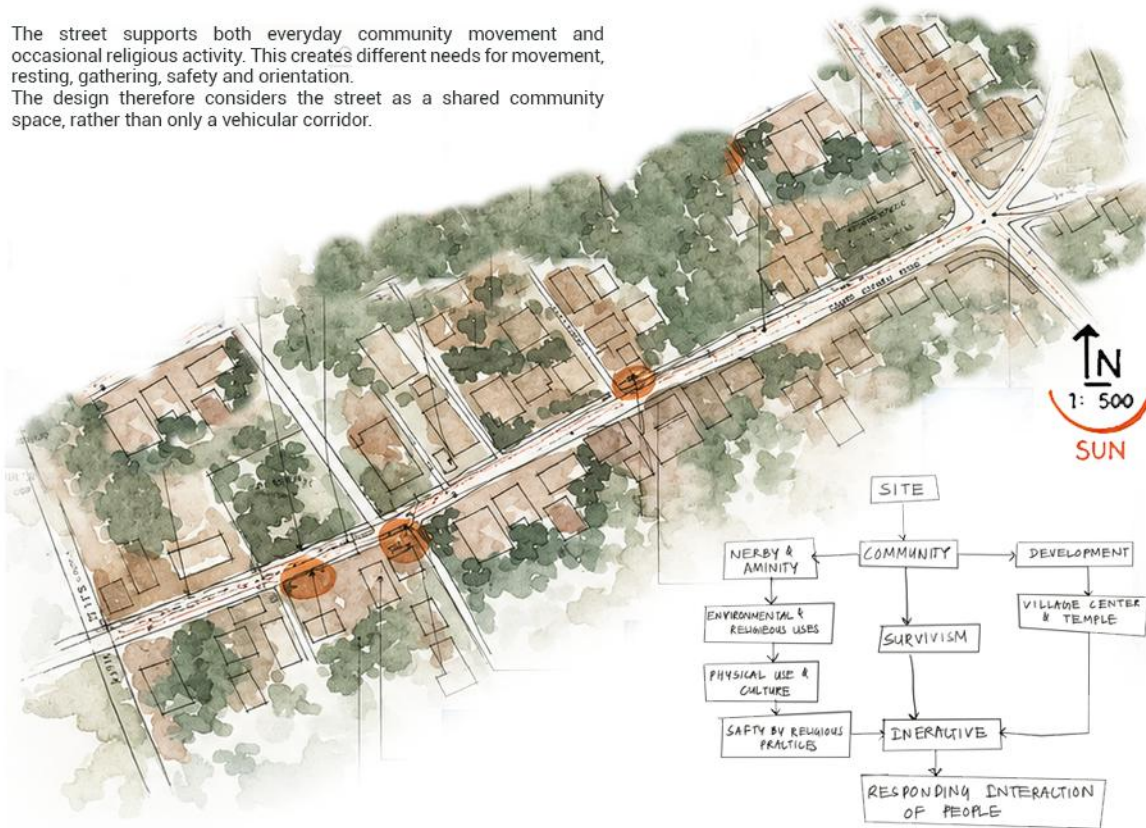
Site observation identified how people move, gather, wait, interact and approach the temple. Existing houses, vegetation, cemetery edges, religious elements and informal spaces already create a strong local character. However, these elements are disconnected and do not currently form a clear public-space system.



Section Q - Q'



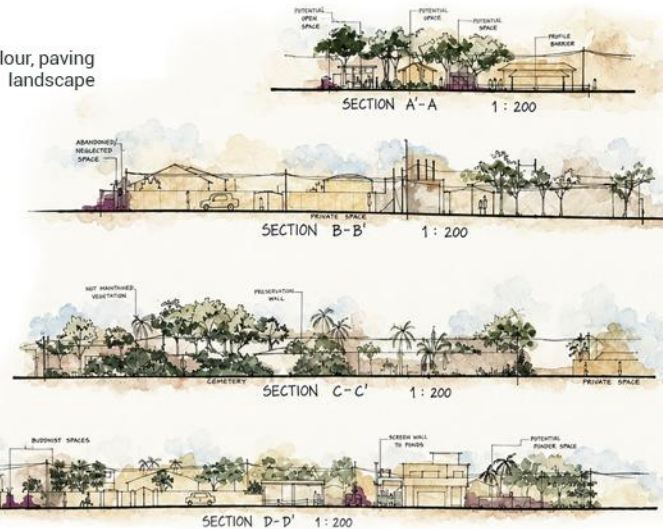
The street supports both everyday community movement and occasional religious activity. This creates different needs for movement, resting, gathering, safety and orientation. The design therefore considers the street as a shared community space, rather than only a vehicular corridor.



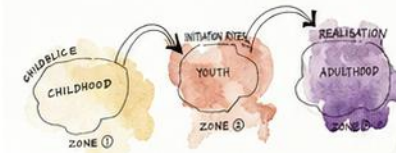
Buddhist cultural references are interpreted through colour, paving patterns, spatial rhythm, planting and selected landscape elements.

The existing street is primarily a linear movement corridor. The design breaks this single line of movement by introducing moments of pause, interaction, planting, colour and cultural expression.

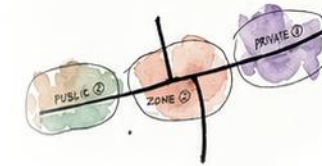
Move - Gather - Experience - Arrive



TIME LINE



ZONING



PUBLIC ENTRANCE (zone 1)

Everyday neighbourhood activities, movement and informal interaction.

COMMUNITY STRIP (zone 2)

Landscape elements, colour, planting and public spaces gradually increase the sense of ownership.

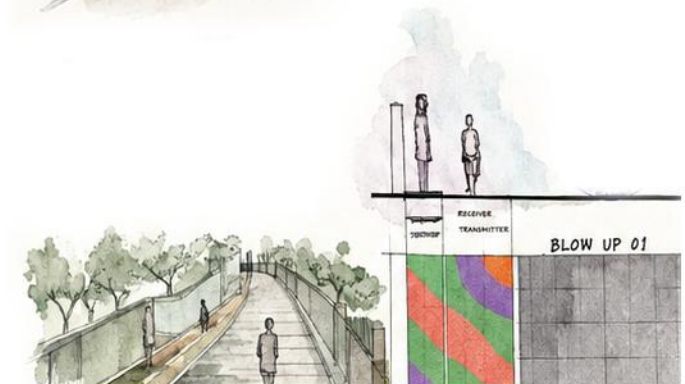
SCARED ARRIVAL STRIP (zone 3)

The landscape becomes calmer and more ordered near the temple, strengthening the visual and cultural relationship with Kande Viharaya.



PUBLIC ENTRANCE (zone 1)

A defined pedestrian route improves safety and accessibility while connecting houses, community spaces and the temple. The walkway creates a stronger relationship between pedestrians and the surrounding neighbourhood.





Design Section N-N'



Design Section M-M'



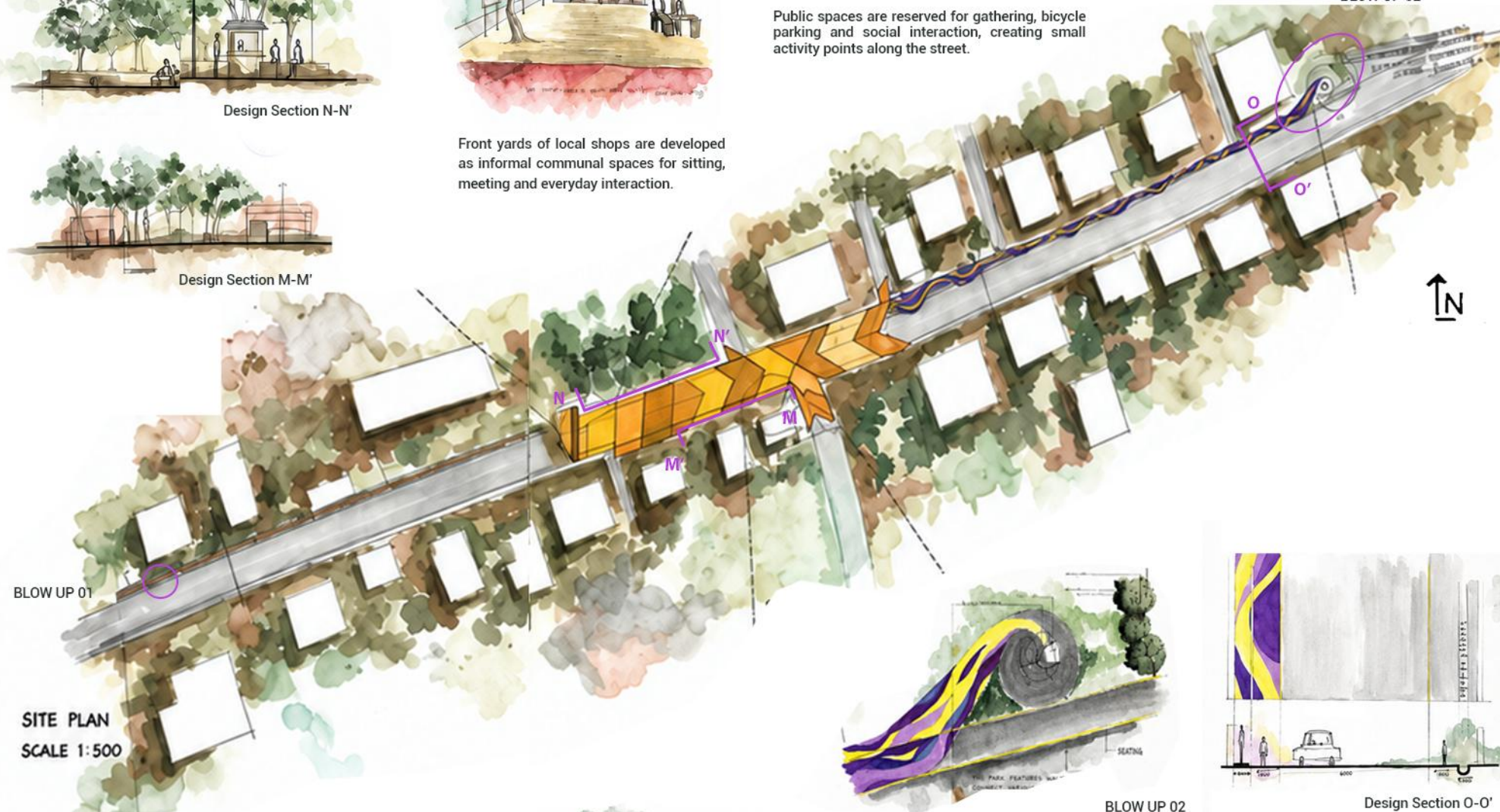
Front yards of local shops are developed as informal communal spaces for sitting, meeting and everyday interaction.

COMMUNITY STRIP (zone 2)

Coloured street elements encourage residents to come outside, interact and experience the street as a shared community space.

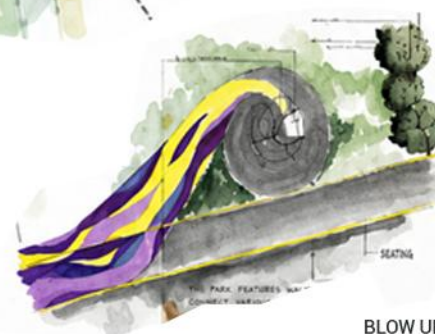
Public spaces are reserved for gathering, bicycle parking and social interaction, creating small activity points along the street.

BLOW UP 02

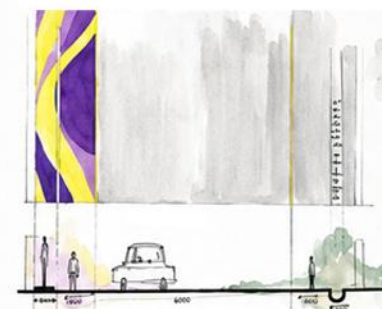


BLOW UP 01

SITE PLAN
SCALE 1:500



BLOW UP 02



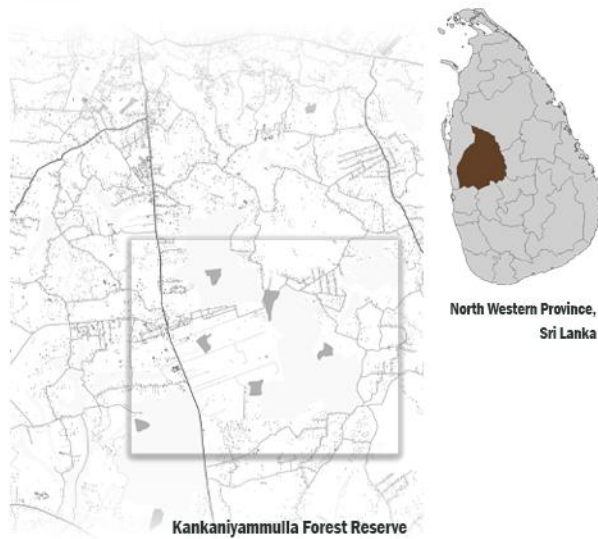
Design Section O-O'



Design Section P-P'

SCARED ARRIVAL STRIP (zone 3)

The pathway continues towards the Buddha statue and temple, creating a clear and continuous cultural journey through the streetscape.



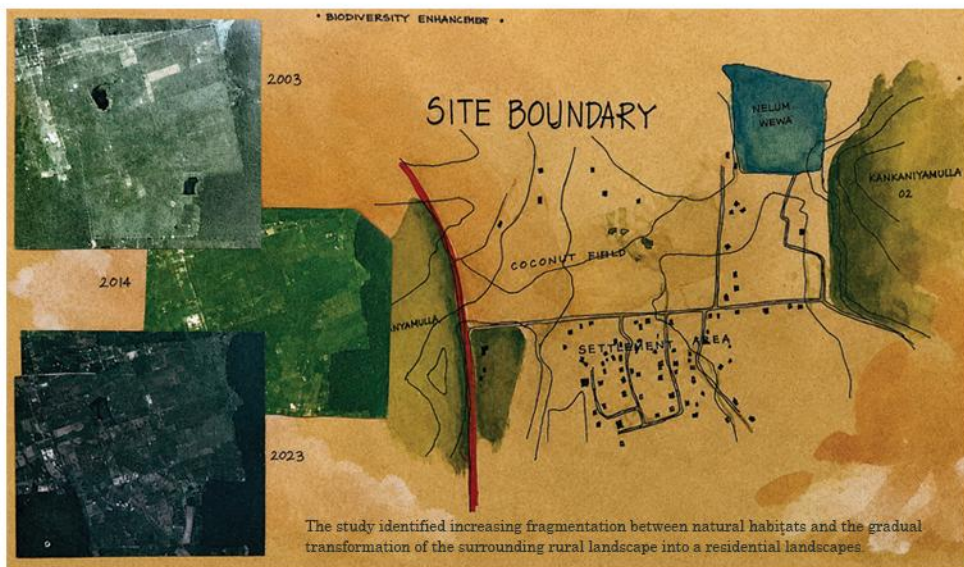
WHISPERS OF DREAMSCAPE

A Firefly-led Ecological Connectivity Landscape at Kankaniyamulla

The project explores how a fragmented rural landscape can be restored as an ecological network while allowing people to experience nature with minimum disturbance.

The site supports five of the seven firefly species identified within Sri Lanka, including endangered species, making it an important area for firefly habitat conservation. Fireflies were therefore selected as the main focal and indicator species, as their presence is closely associated with healthy, moist and relatively undisturbed environmental conditions. Their habitat requirements guide the project's ecological restoration strategy, reconnecting forest, wetland, paddy and plantation landscapes through habitat enhancement, low-impact access and carefully controlled visitor experiences.





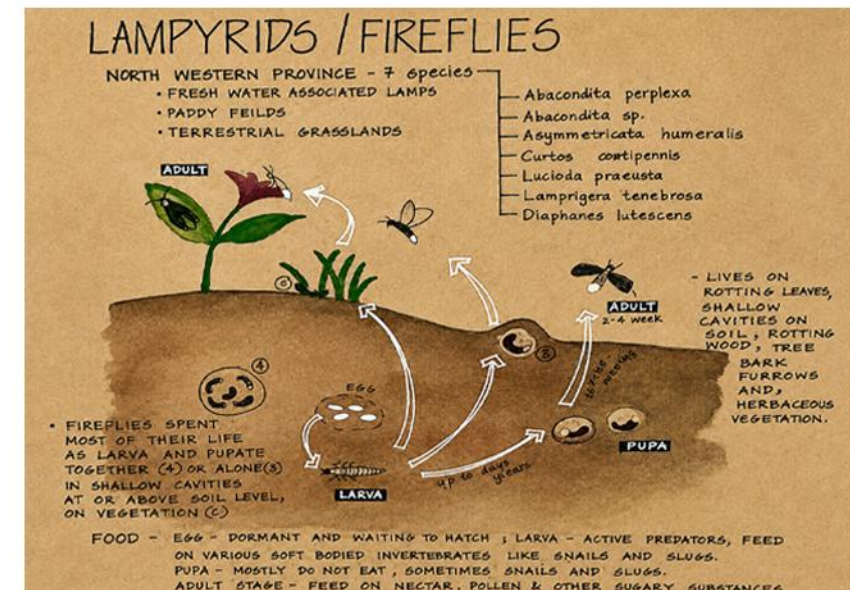
ECO-SYSTEMS

	FOREST	PADDY / COCONUT	WATER TANK	HOMESTEAD
	WET, DENSE MOSTLY OPEN	DENSE, SHADED HIGH	SEMI DENSE, MIXED MULTI LAYERED	
FLORA	<i>Swietenia macrophylla</i> - MAHOGANY <i>Xylocarpus parviflorus</i> - NATANI <i>Artocarpus nobilis</i> - WILD DEL <i>Dipterocarpus zeylanicus</i> - HORA <i>Mangifera zeylanica</i> - TAMBA <i>Canarium umbro</i> - KITMUL <i>Terminalia arjuna</i> - KUMBUK <i>Dalbergia retusa</i> - GODAPARA <i>Syzygium wallia</i> - WALLA PATTI <i>Macaranga peltata</i> - KAMPA	<i>Oryza sativa</i> - RICE <i>Cocos nucifera</i> - COCONUT <i>Carica papaya</i> - PAPAYA <i>Piper betle</i> - BETEL <i>Tamarindus indica</i> - TAMARIND	<i>Vernonia cinerea</i> - NONNEADUDUMBARA <i>Eichhornia crassipes</i> - JAPAN JABARA <i>Sida rhombifolia</i> - BABILA <i>Commelina diffusa</i> - GIRA PENELA <i>Cardiospermum halicacabum</i> - WEL PENELA <i>Osebeckia octandra</i> - KEEN BOVITIYA	<i>Passiflora edulis</i> - PASSION <i>Asadirachta indica</i> - KINOMBA <i>Manihot esculenta crantz</i> - MANIOC <i>Moringa oleifera lam.</i> - MURUNGA
FAUNA	<i>Narco sinica</i> - RILAWA <i>Gallus lafayettii</i> - WANA KUKULA <i>Arctopavo conensis</i> - MONARA <i>Loniculus berullinus</i> - PARROT <i>Haliotus leucospeneus</i> - HANIK	<i>Loniculus berullinus</i> - PARROT <i>Naja naja</i> - CORRA <i>Bos taurus</i> - COW <i>Arctopavo conensis</i> - MONARA	Amphibians Reptiles Invertebrates	<i>Gallus cinica</i> - WALI <i>Loniculus berullinus</i> - KUKULA <i>Felis catus</i> - CAT <i>Canis lupus familiaris</i> - DOG <i>Bos taurus</i> - COW <i>Gallus domestic</i> - CHICKEN

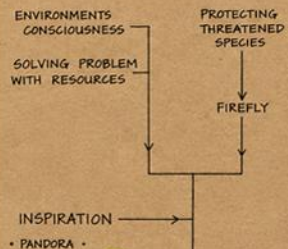
Existing ecological assets - forest cover, Nelum Wewa, small ponds, drainage lines, paddy systems and mature vegetation.
 Existing pressures - habitat fragmentation, settlement expansion, agricultural change, human disturbance and uncontrolled access.

Existing opportunities - abandoned agricultural land, strong water systems, mature vegetation, wildlife movement and existing firefly populations.

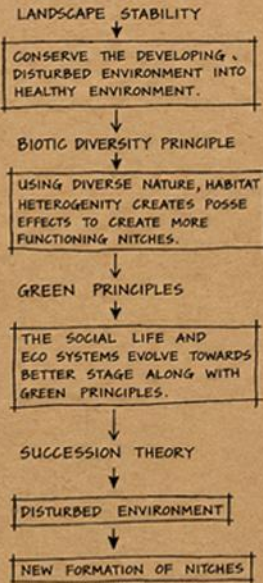
The site already contains a strong combination of forest edge, moist soil, standing water, paddy landscapes, decomposing vegetation and shaded conditions. These overlapping habitats create an ecologically rich transition zone capable of supporting insects, amphibians, birds and other wildlife.



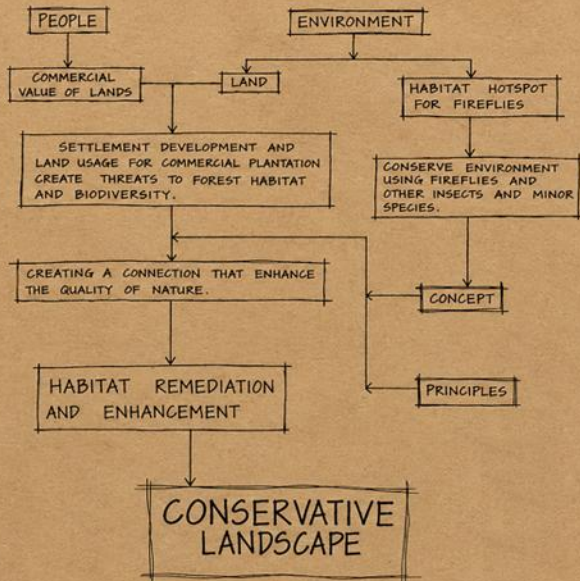
CONCEPTUAL APPROACH



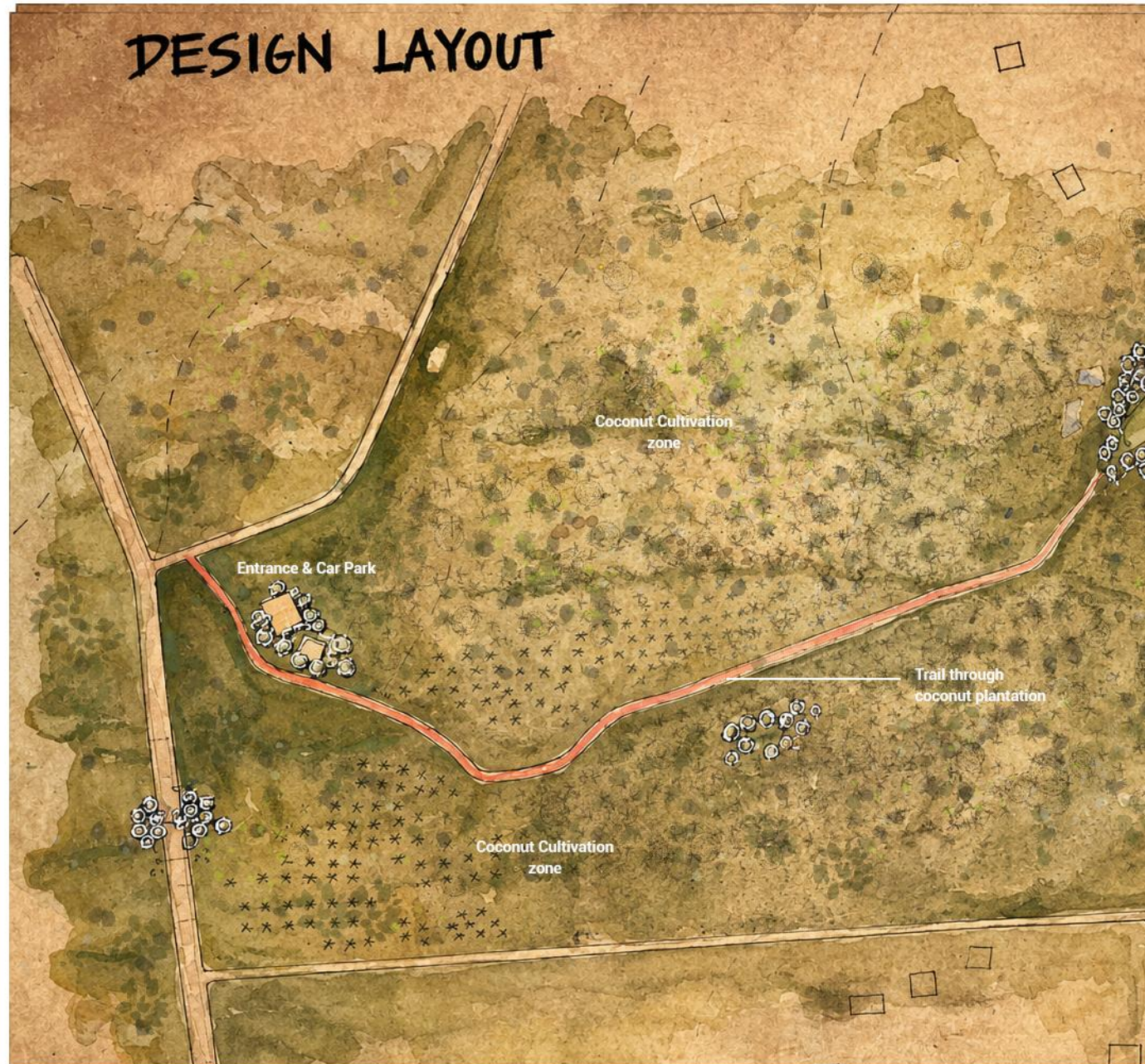
THEORATICAL APPROACH

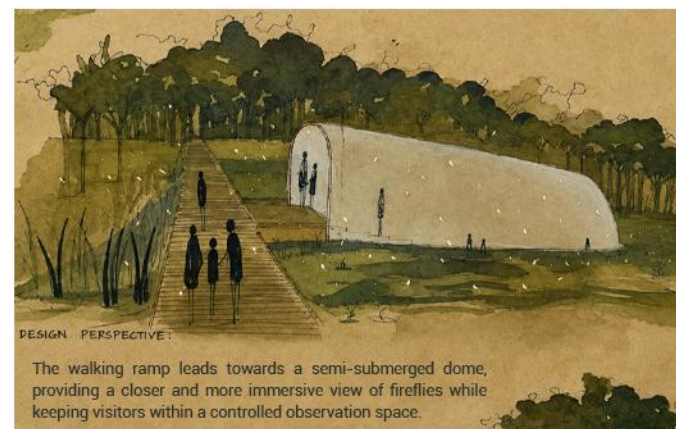
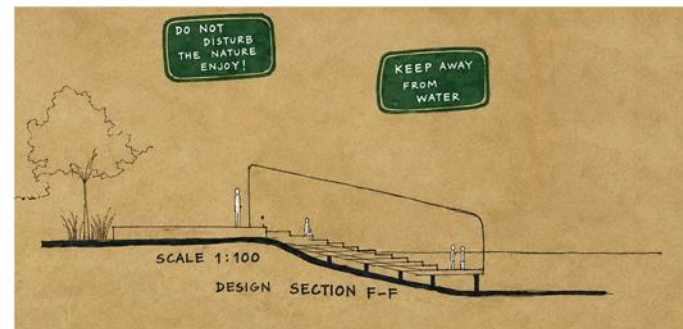


PROJECT APPROACH



DESIGN LAYOUT





Boardwalk: Low-impact trail for quiet firefly viewing.

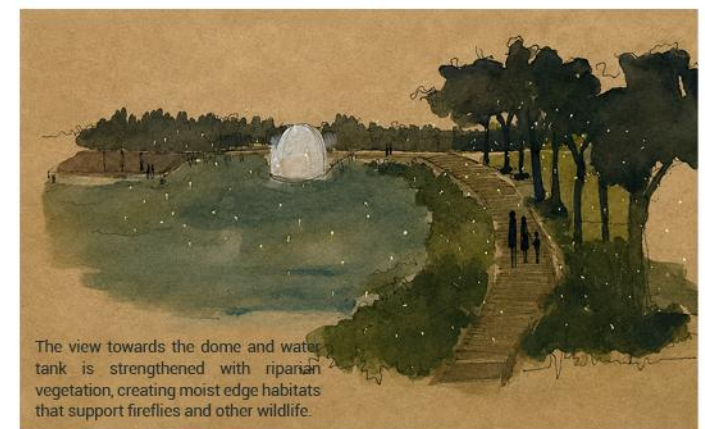
Lighting: Minimal, low-intensity lighting protects dark habitat and firefly communication.

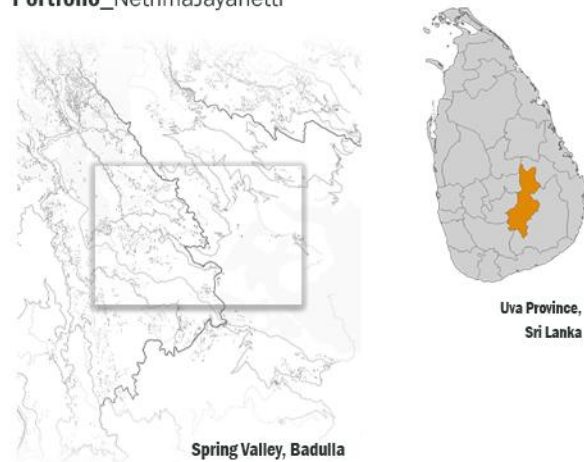
Observation Dome: A controlled viewing point overlooking the wetland and firefly habitat.

Pond Edge: Protected ecological core strengthened with riparian planting.

Campsite: Low-impact nature stay with limited disturbance and minimal structures.

Planting: Multilayered vegetation supports shade, moisture, food and habitat connectivity.





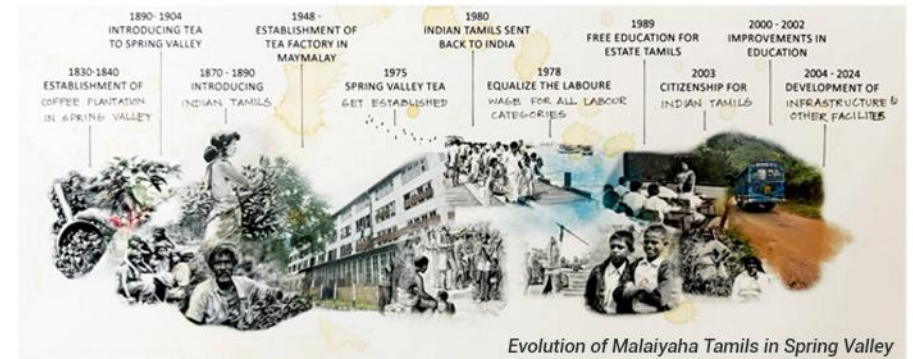
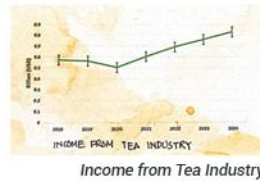
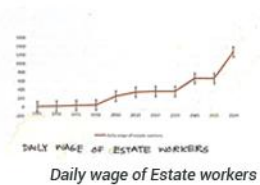
FROM ROOTS TO RADIANCE

The Journey of Growth and Identity of Spring Valley, Badulla, Sri Lanka.

Located within the tea-estate landscape of Spring Valley, Badulla, this project explores how landscape architecture can support the growth, recognition, and empowerment of the Malaiyaha Tamil community.

The project responds to the community's historical social, spatial, and economic marginalization, and the undervaluation of Malaiyaha Tamil culture. Although this community has played a significant role in Sri Lanka's tea industry, its identity, heritage, and contribution remain insufficiently acknowledged.

The design therefore proposes an interconnected network of cultural, social, educational, economic, and recreational spaces integrated into the existing terrain and tea landscape. Through spatial experiences, the landscape becomes a platform for strengthening cultural pride, changing attitudes, improving quality of life, and promoting positive social integration.

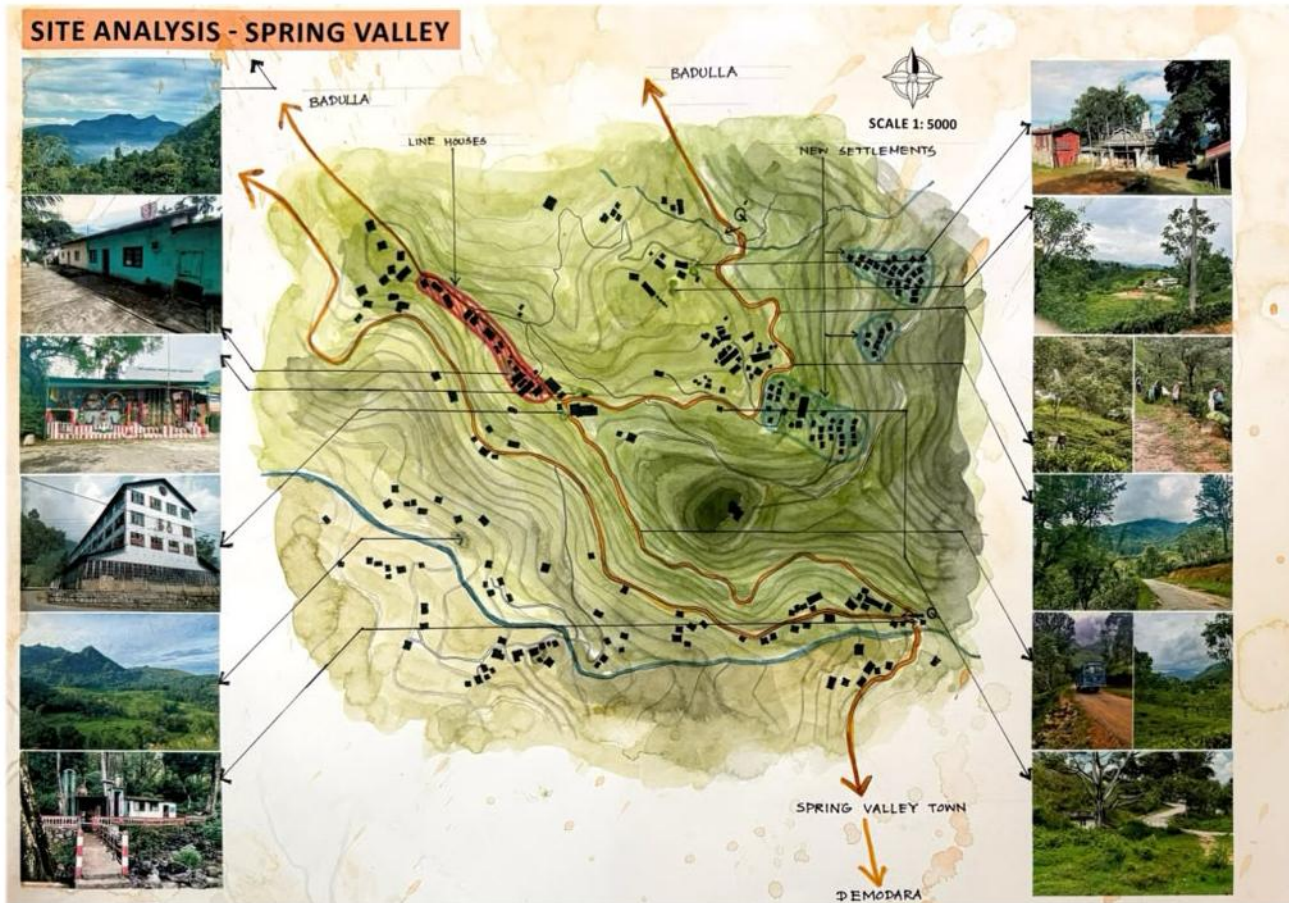


Tea is a globally recognized symbol of Sri Lanka, with Ceylon Tea celebrated for its distinctive flavour, exceptional quality, and contribution to the country's economy and cultural identity. This international reputation was built over more than a century through labour-intensive cultivation methods, largely sustained by plantation workers whose skill, precision, and endurance shaped the tea industry.

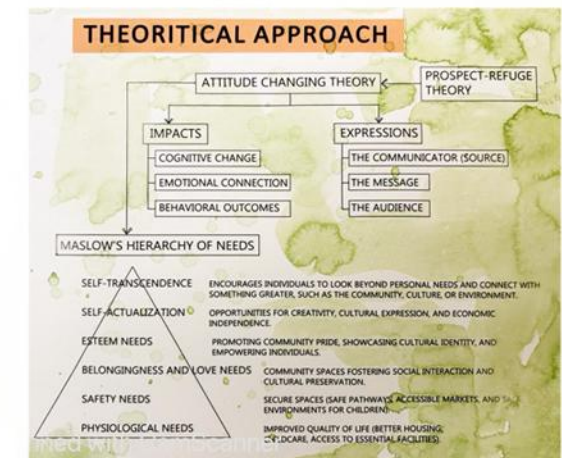
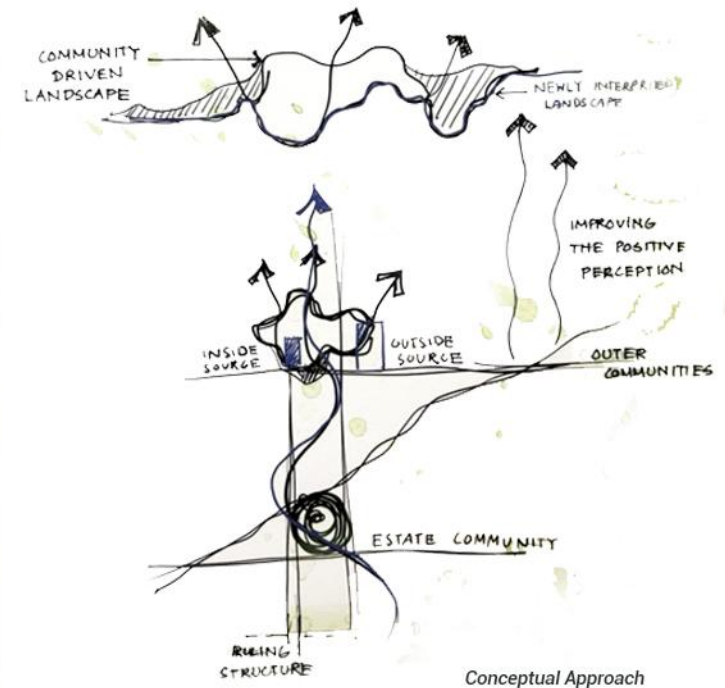


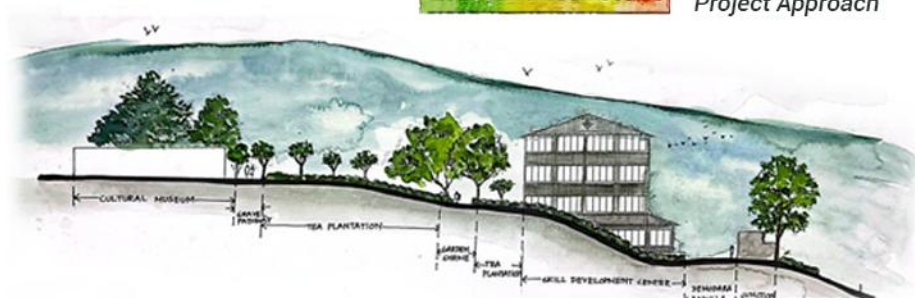
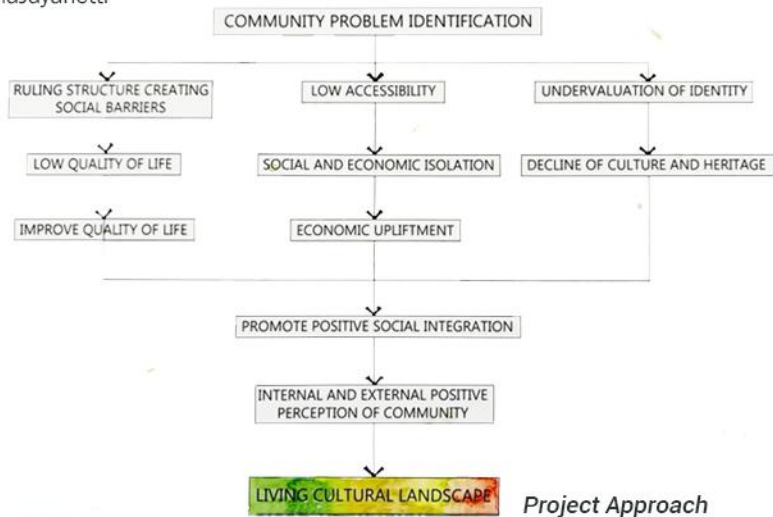
During British colonial rule, Tamil labourers were brought from South India to work on Sri Lanka's tea estates. Despite their essential contribution, they endured long working hours, low wages, inadequate housing, and limited access to education, healthcare, citizenship, and social mobility.

Although generations of Malaiyaha Tamil workers have played a central role in establishing the identity and global value of Sri Lankan tea, their cultural heritage, lived experiences, and contribution remain insufficiently recognized. **This historical contradiction—between the celebrated image of Ceylon Tea and the marginalized lives of the people who sustained it—forms the critical foundation of the project.**



The project identifies three interconnected community issues:
 Social barriers created by the historical estate-management structure
 Limited physical and social accessibility
 Undervaluation of Malaiyaha Tamil identity, culture, and heritage





The neglected tea factory is adaptively reused as a skill development centre, giving new purpose to an existing heritage structure. An adjoining tea museum presents the history of the estate, tea culture and the everyday life of the community, connecting local identity with learning and visitor experience.



Located beside the skill development centre, the open-air theatre creates a shared space for learning, performances, dancing and community gatherings within the tea landscape.

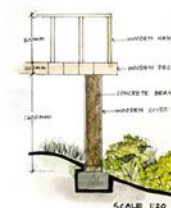


A previously neglected ground around the workers' shrine is transformed into a calm, nature-integrated spiritual space, connecting everyday working life with cultural and religious practices.





A sacred cultural space surrounding the community Kovil, designed to support worship, gathering and cultural continuity while strengthening the identity of the Spring Valley community.



An elevated trail connects the tea plantations through low-impact pathways, offering viewing points for visitors and resting spaces for tea pluckers.



On Festival Days



On Normal Days



Flexible market stalls support everyday trading and transform during cultural events. Vibrant colours inspired by Malayalam Tamil culture strengthen the visual identity and festive character of the space.



Located along the main road and close to the bus stop, the milk collection centre is an active morning gathering point where locals bring milk and schoolchildren can enjoy a fresh cup before travelling to school



A safe play area within the site allowing parents working in the surrounding plantations to remain visually connected with them.



Western Province,
Sri Lanka

Karadiyana Landfill, Piliyandala, Sri Lanka.

REGROUNDING THE LANDFILL

Transforming Waste Ground into a Resilient Landscape

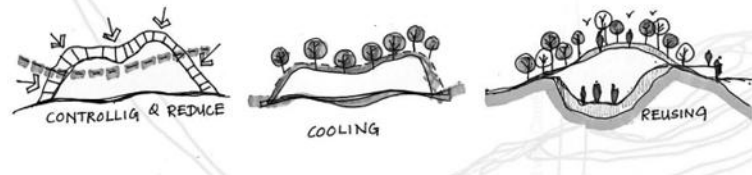
Landfills today represent a critical environmental challenge within rapidly urbanizing cities. They function not only as waste disposal sites but also as zones of heat accumulation, ecological degradation, and environmental stress. These landscapes often disrupt natural systems and create a clear separation between people and their environment.

The project repositions the landfill as a performative learning landscape, a community forest, where its existing identity is reinterpreted to build awareness about waste systems, ecological processes, and human responsibility.

In the long term, the landscape is designed to evolve as a self-sustaining ecological system within a rapidly urbanizing context. It functions as a cooling environmental core. The project ultimately establishes that landfill sites can shift from zones of environmental stress into resilient, productive landscapes that contribute to both ecological stability and urban life.



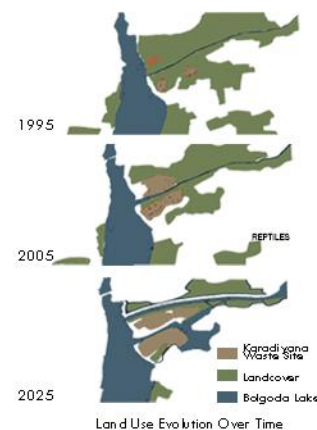
Reduce – Cool – Reuse



Reduce the negative impacts of the landfill by addressing heat accumulation, emissions, and water-related issues, creating a cleaner, and more stable environment.

Rebuild the landfill as a functioning and **cooling** habitat system by supporting ecological succession and improving biodiversity through diverse and interconnected landscapes.

Reuse the landfill into a socially inclusive landscape that encourages public engagement, improves environmental perception, and fosters responsibility through experience and interaction.

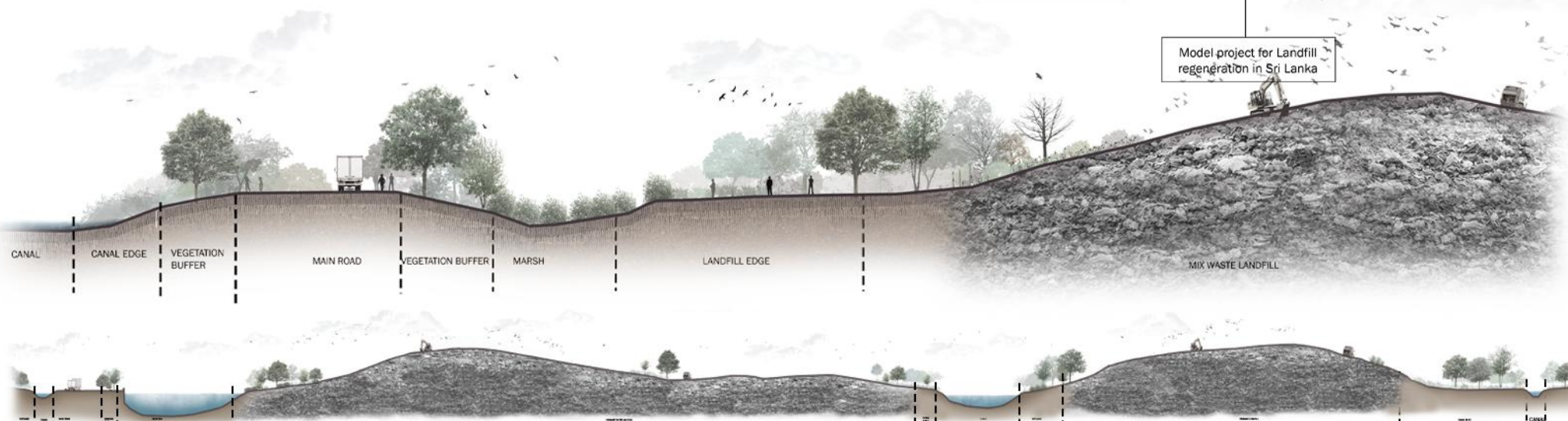
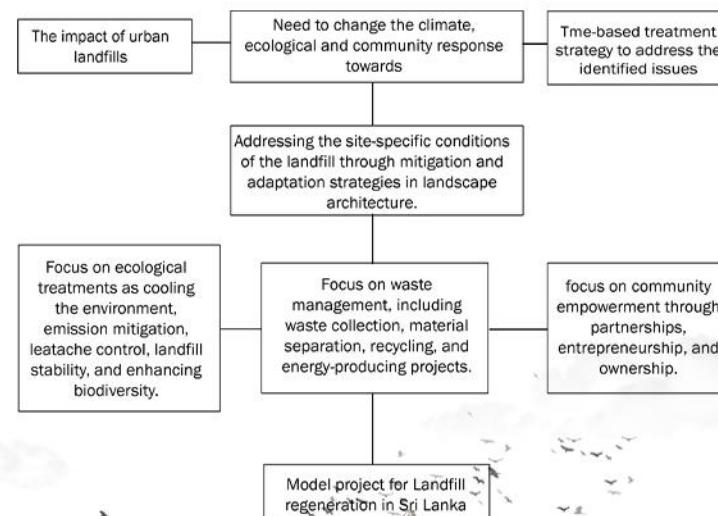


The diagram illustrates the pathways of contaminants from a landfill to various environmental compartments and organisms. The central element is a **LANDFILL** mound, which is the source of **LANDFILL GASES, FUMES, SMOKE AND DUST**. Arrows show the following pathways:

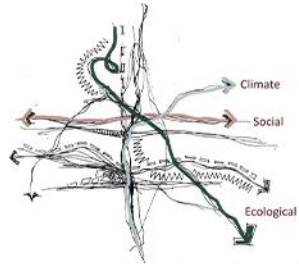
- Contaminated Surface:** Arrows point from the landfill to the ground surface, which is labeled **CONTAMINATED SURFACE**.
- Soil and Groundwater:** Arrows show contaminants moving from the surface into the **SOIL** and then into the **GROUNDWATER** table.
- Leachate Pathways:** Arrows show **LEACHATE PATHWAYS** moving from the landfill into the **SOIL** and **GROUNDWATER**.
- Solid Waste:** Arrows show **SOLID WASTE** moving from the landfill into the **SOIL**.
- Biota:** Arrows show contaminants moving from the **CONTAMINATED SURFACE** to **PLANTS AND ANIMALS** (represented by a bird and a fish). The bird is labeled **AVIFAUNA** and the fish is labeled **FISH**.
- Atmosphere:** Arrows show contaminants moving from the landfill into the **ATMOSPHERE** (represented by a cloud) and then to **CONTAMINATED RIVER SEDIMENTS** and **CONTAMINATED RIVER FOOD CHAINS**.
- Rain Deposition:** Arrows show **RAIN DEPOSITION** moving from the atmosphere to the **CONTAMINATED SURFACE** and **CONTAMINATED RIVER SEDIMENTS**.
- Mammals:** An arrow shows contaminants moving from the landfill to **MAMMALS** (represented by a mouse).

A legend on the right side of the diagram identifies the symbols used:

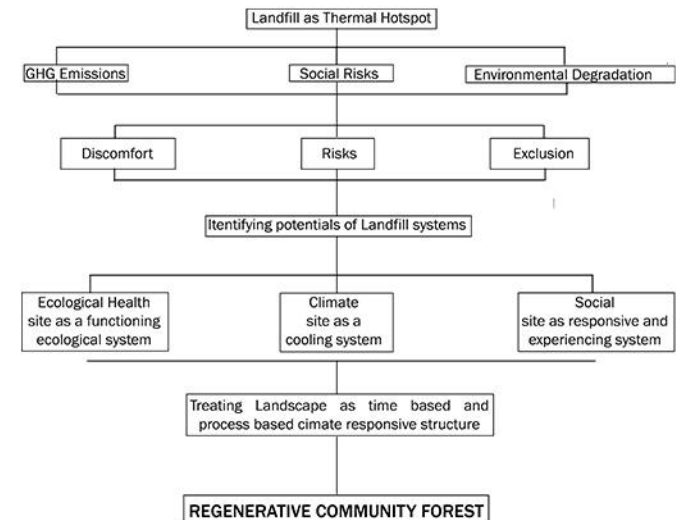
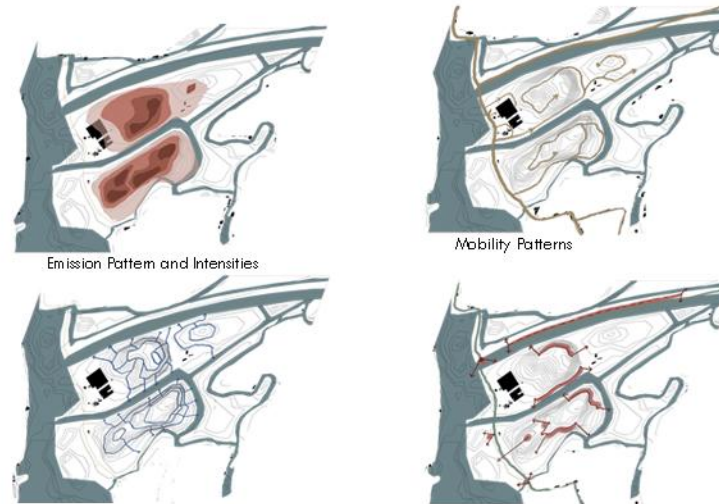
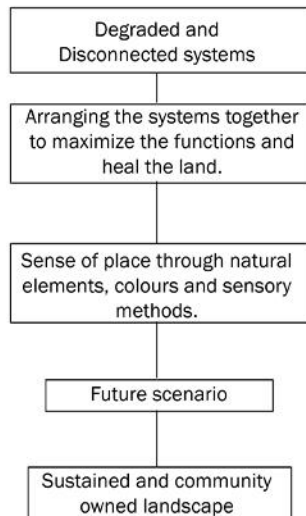
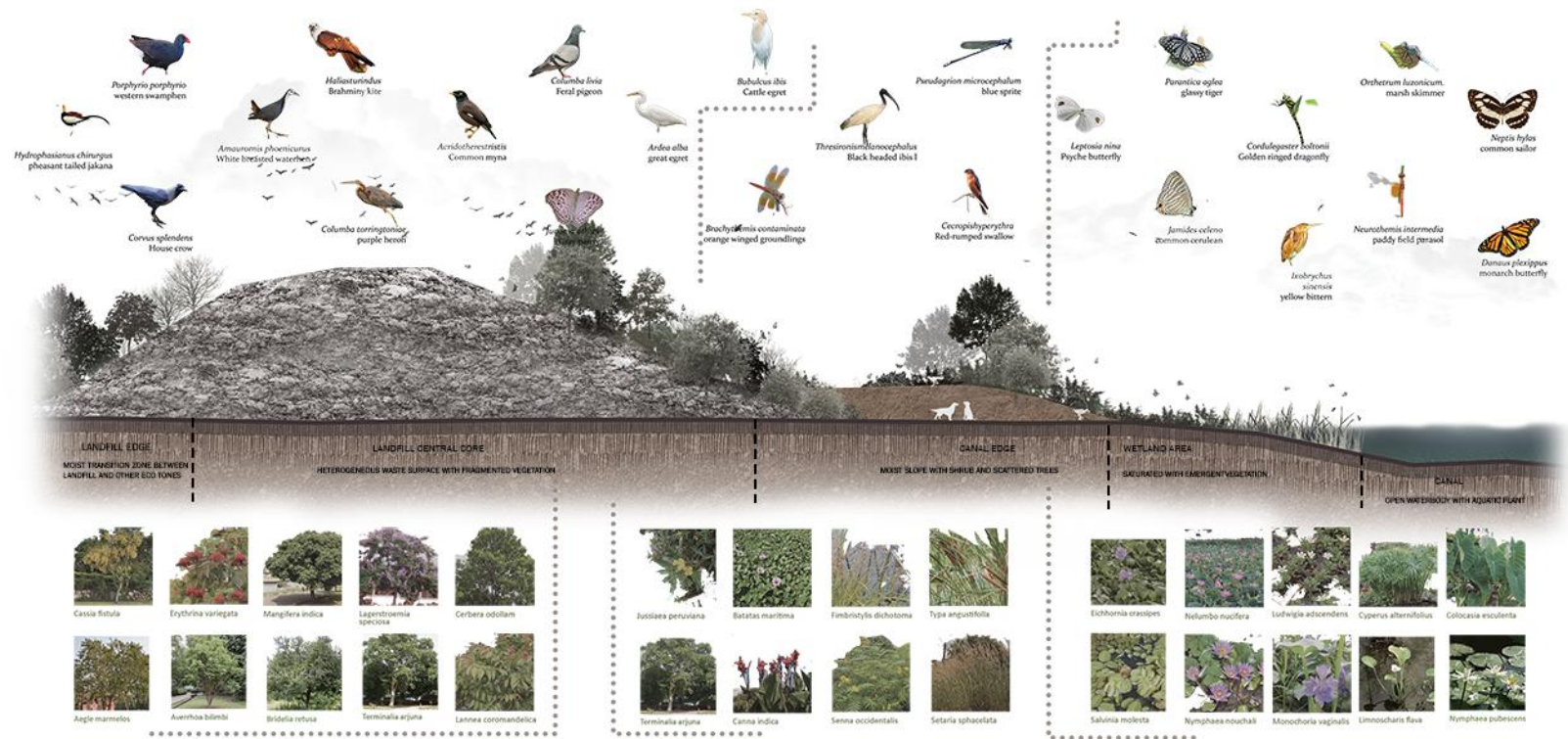
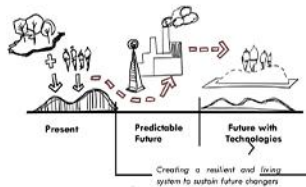
- CONTAMINANTS THROUGH FOOD CHAINS** (represented by a red arrow)
- CONTAMINATED RIVER SEDIMENTS** (represented by a red arrow)
- CONTAMINATED RIVER SEDIMENTS** (represented by a blue arrow)
- GROUNDWATER TABLE** (represented by a blue arrow)
- LEACHATE PATHWAYS** (represented by a blue arrow)
- SOLID WASTE** (represented by a brown arrow)
- RAIN DEPOSITION** (represented by a red arrow)

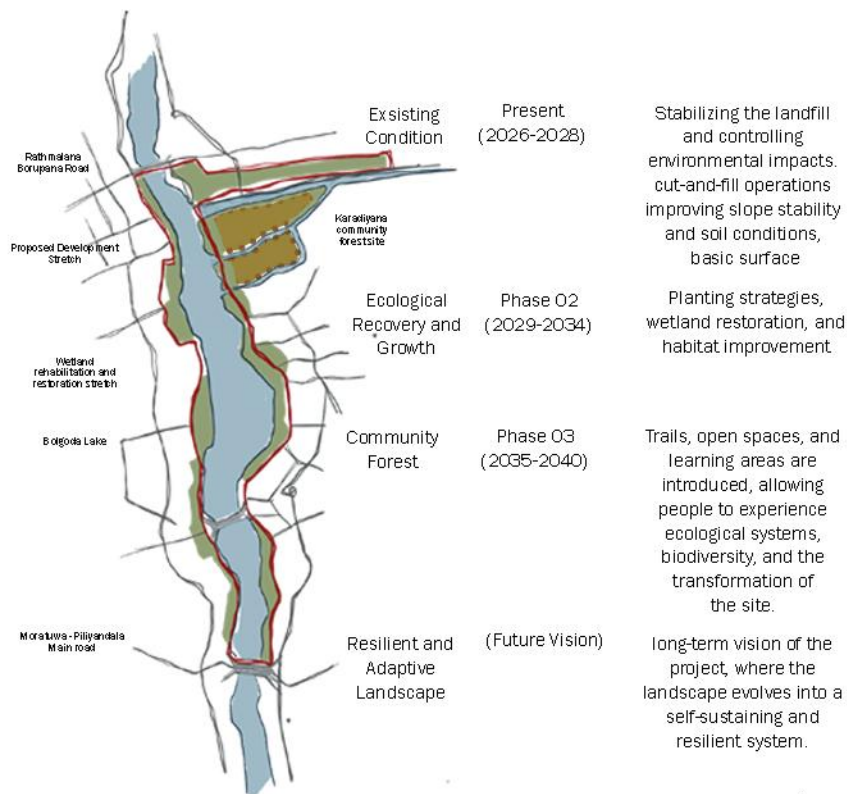


Regrading the Mosaics



The concept of "Regrading the Mosaics" envisions the landscape as a resilient and adaptive system that can sustain ecological processes and support community use over time. The framework also allows flexibility for future changes, including technological advancements and evolving environmental conditions.





Stabilizing the landfill and controlling environmental impacts. cut-and-fill operations improving slope stability and soil conditions, basic surface

Early engagement of surrounding communities. Opportunities are created for participate as workers and contributors in waste management and site improvement activities.

Planting strategies, wetland restoration, and habitat improvement.

The community continues to play an active role dby participating in maintenance, planting, and monitoring activities.

Trails, open spaces, and learning areas are introduced, allowing people to experience ecological systems, biodiversity, and the transformation of the site.

Community members become part of the management and operational structure.

long-term vision of the project, where the landscape evolves into a self-sustaining and resilient system.

Strong community ownership ensures continuous care and adaptation.

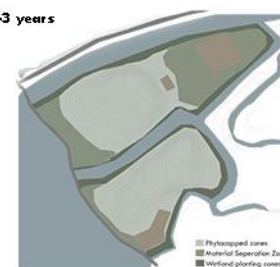
Rigid edges indicate stopping and aligned spatial arrangements

Pausing and reviving nodes

Organic wayforwards to guide in a free forms



0-3 years



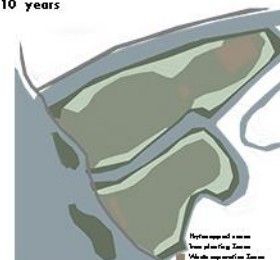
05 - Regrounding the Landfill

Organized Material Separation Zone

Installation of emission pipes

Landfill stabilization by CUT and Fill

3-10 years



Phytocapping for ecological recovery

Planting forest trees with root capsules

Wetland plantation as natural drainage system

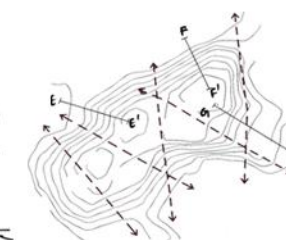
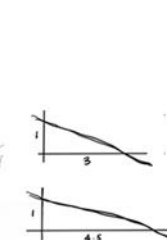
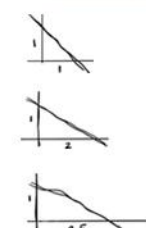
10+ years

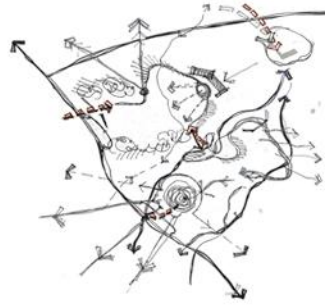


Secondary forest patch as the ecological habitat

Community Responsive and engaging spaces

Differential settlement gradually reduces sharp contour variations
high settlement (~50-150 mm/year)
Late: minimal settlement (<20 mm/year)





ZONE A: BIOLOGICAL RESEARCH & BIODIVERSITY EXPERIENCE

A - Arrival Frame



C - Learning Grounds



B - The Habitat Trail



D - KAYAK DECK



ZONE B: COMMUNAL ENGAGEMENT & RECREATION

E - FREEDOM LAWN



G - VALLEY OF SCULPTURES



F - WATER EDGE WALK



H - SKYLINE HILL

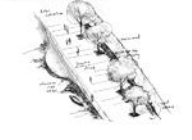


I - SHARED STRIP

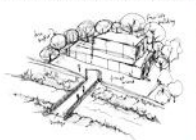


ZONE C: RECYCLING & CIRCULAR METABOLISM ZONE

J - SETTLING EDGE



K - WASTE MANAGEMENT FACILITY



1. Entrance
2. Forest Trail
3. Research Center
4. Freedom Grounds
5. Wetland trail
6. SWaste Sculpture Lands
7. Urban view Hills
8. Community Market
9. Waste Faacility
10. Roadside decks

THE HABITAT TRAIL

The Habitat Trail is designed as a living edge where forest, meadow, wetland, and viewing platforms gradually meet. The landscape shifts softly through landform, vegetation, and water, allowing people to experience the ecological richness of the regenerated landfill.

This space works like a series of framed habitat rooms. Dense planting opens into small glades, shaded paths lead into sunlit clearings, and wetland edges reveal birds, insects, grasses, flowers, and water movement.



TREATED WOOD



GREEN CONCRETE WITH SUPPLEMENTARY CEMENTITIOUS MATERIALS



6'x12' CONCRETE CURB

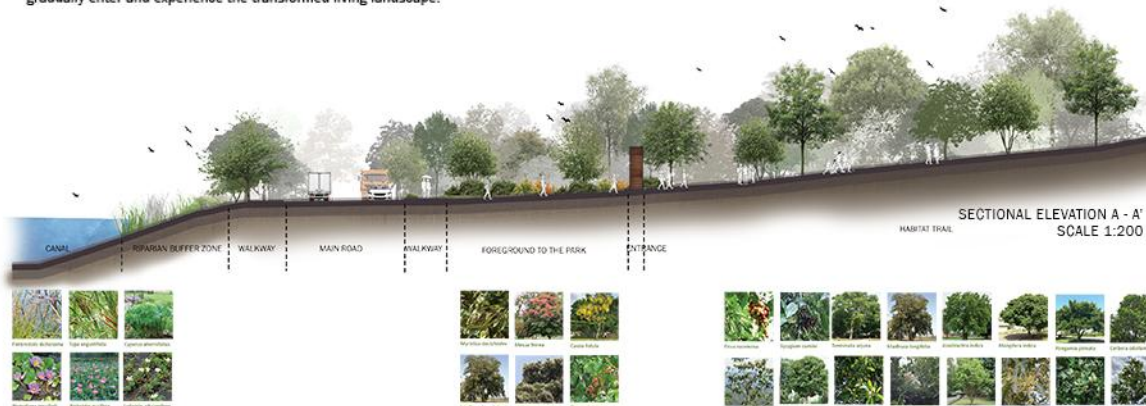
ARRIVAL FRAME

The Community Forest entrance forms the first threshold between landfill memory and regenerated ecological mosaic. Acting as a landscape frame, it marks arrival while revealing only glimpses of tree trunks, filtered light, shaded paths, and vegetation. This creates mystery, inviting visitors to gradually enter and experience the transformed living landscape.



LEARNING GROUNDS

The Research Centre is oriented around the Learning Grounds, a shared courtyard where landfill memory, research activity, and community education come together. At its core, memorial notice boards trace the different time phases of the landfill, allowing visitors and researchers to understand the site's transformation from waste ground to regenerative landscape.

SECTIONAL ELEVATION A - A'
SCALE 1:200SECTIONAL ELEVATION C - C'
SCALE 1:200

Beside this interpretive edge, an open research yard provides a quieter space for researchers to pause, discuss, observe, and engage in informal activities. The ground, designed with amphitheatre, shaded edges, create a calm atmosphere, turning the centre into a place of personal living landscape.



CONCRETE PAVEMENT

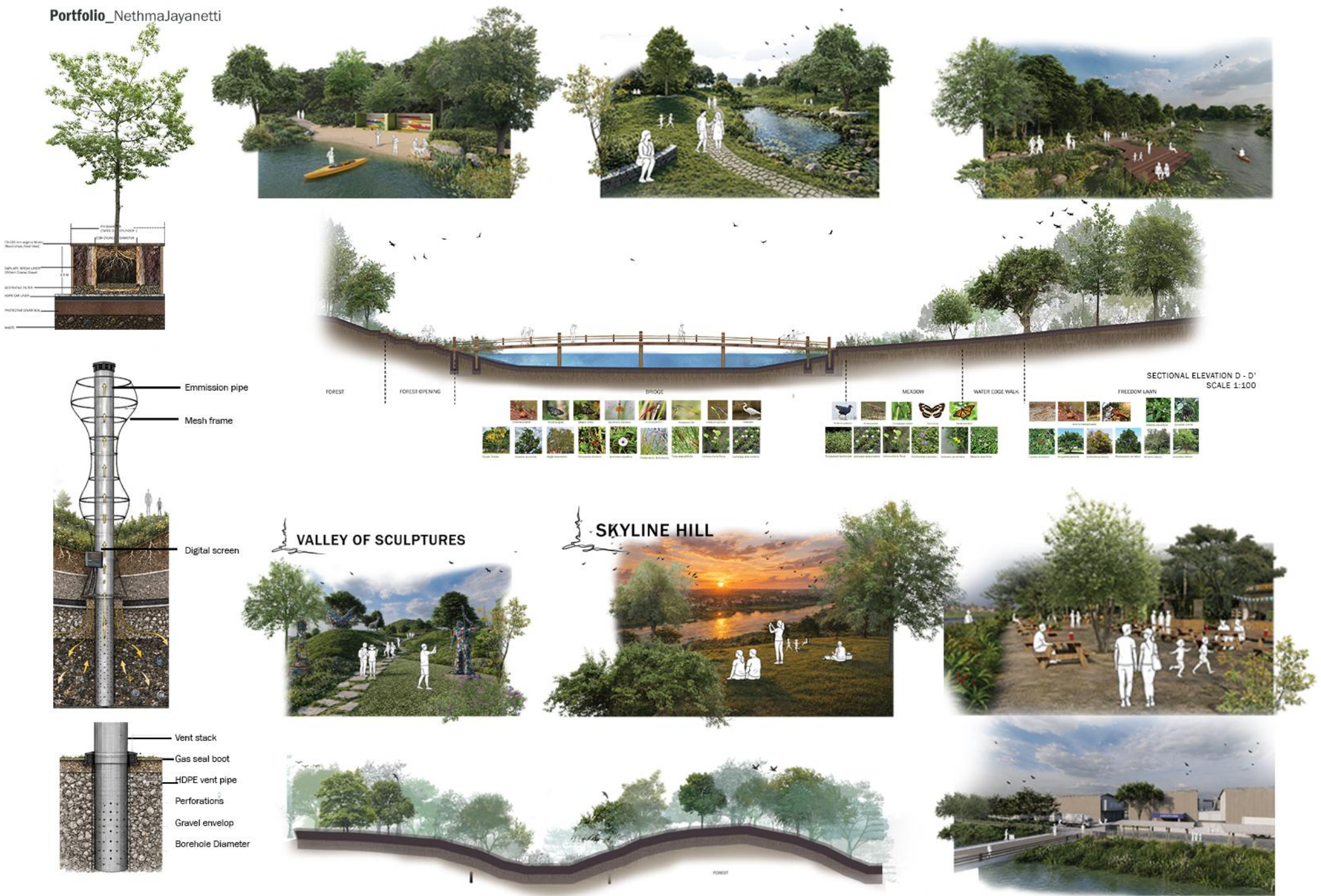


RECYCLED METAL



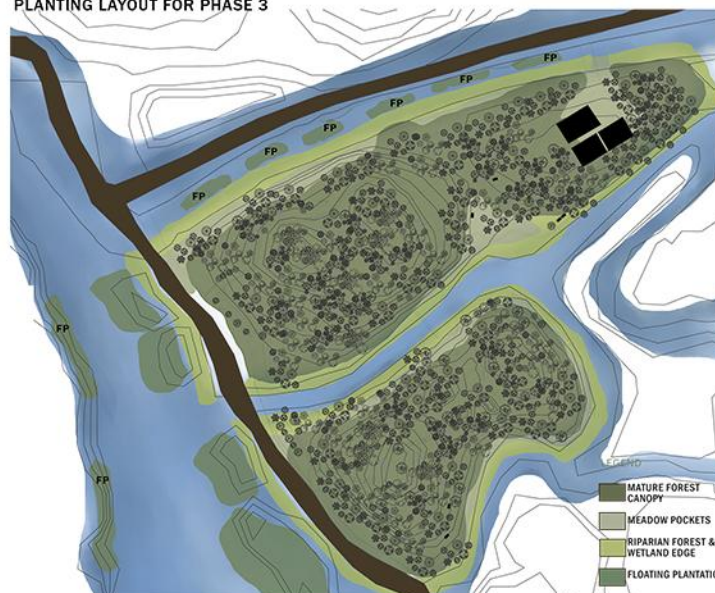
VEGETATED CONCRETE





PLANTING PLAN

PLANTING LAYOUT FOR PHASE 3



Kumbuk <i>Terminalia arjuna</i>	Munamal <i>Mimusops elengi</i>
Murutha <i>Lagerstroemia speciosa</i>	Godapara <i>Dillenia retusa</i>
Kohomba <i>Azadirachta indica</i>	Kebella <i>Careya arborea</i>
Mee <i>Madhuca longifolia</i>	Domba <i>Calophyllum inophyllum</i>
Magul Karada <i>Careya arborea</i>	Anoda <i>Annona squamosa</i>
Gansuriya <i>Senna surattensis</i>	Belli <i>Aegle marmelos</i>
Gon Suriya <i>Thespesia populnea</i>	Maara <i>Samanea saman</i>
Malaboda <i>Hibiscus tiliaceus</i>	Bilin <i>Averrhoa bilimbi</i>
Naa <i>Mesua ferrea</i>	Ketakela <i>Pandanus odorifer</i>
Mango <i>Mangifera indica</i>	Waralu <i>Eleocharis serratus</i>
Ehela <i>Cassia fistula</i>	Milla <i>Vitex pinnata</i>
Kenda <i>Macaranga peltata</i>	Koon <i>Scheuchzeria oleosa</i>
Erabadu <i>Erythrina variegata</i>	Ruk Attana <i>Astonia scholaris</i>
Atthikkka <i>Ficus racemosa</i>	Madan <i>Syzygium cumini</i>

The planting concept is based on a gradual ecological transformation where the landfill surface is not planted as a finished forest at once, but guided through controlled succession. The landscape begins as a protected phytocapping surface, then develops into a young woodland mosaic, and finally matures into a stable native forest and wetland system.

1. Matured Forest Canopies



2. Matured Riparian and Wetland Forest

The matured riparian and wetland forest strengthens the ecological edge between the landfill and the water system. Species such as *Terminalia arjuna*, *Ficus hispida*, *Ficus racemosa*, *Bridelia retusa*, *Syzygium cumini*, and wetland plants such as *Cyperus alternifolius*, *Eleocharis dulcis*, *Fimbristylis dichotoma*, *Typha angustifolia*, *Nelumbo nucifera*, and *Nymphaea nouchali* are used to create a layered water-edge habitat.

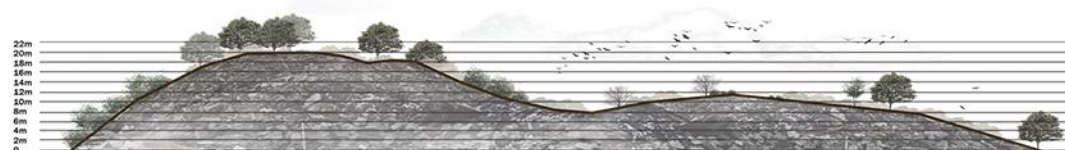
3. Permanent Meadows

Permanent meadows are retained within the mature forest mosaic to prevent the landscape from becoming completely closed and visually dense. Meadow species such as *Melastoma malabathricum*, *Osebeckia octandra*, *Tephrosia purpurea*, *Cynanthium cinereum*, *Aerva lanata*, and controlled ornamental pockets of *Canna indica* can support pollinators and maintain seasonal colour.



PRESENT CONDITIONS

The Karadiyana landfill has reached its functional capacity, having exceeded its intended operational threshold around 2010 and continuing to receive waste for over a decade beyond that point. This prolonged overloading has transformed the site into a highly unstable, unmanaged open dumping system.



20 YEARS AFTER WITHOUT ANY WASTE MANAGEMENT PRACTICES

Over the next 20 years, the landfill will transform into a high-risk environmental hazard zone, amplifying climate impacts, ecological degradation, and human vulnerability, ultimately becoming an irreversible urban wastescape.

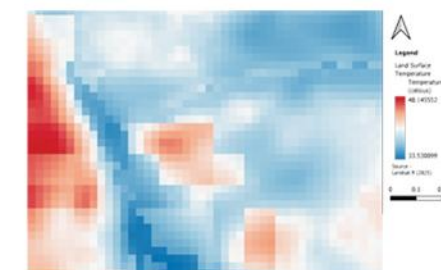


20 YEARS AFTER THE DESIGN IMPLEMENTATION

the proposed regenerative intervention reconfigures the landfill as a controlled ecological system, reducing its physical mass, stabilizing its environmental processes, and transforming it into a cooling, biodiverse, and socially productive landscape.



Land Surface Temperature (LST) - 2025



Land Surface Temperature (LST) - 2050

